

C-DAC — ASSIGNMENT 01 ANSWERS

QUESTION 01

Write a Java program to declare two integer variables and perform addition, subtraction, multiplication, division, and remainder operations. Display all the results.

LOGIC

Create `Scanner` instance. Read `num1` and `num2` from console using `sc.nextInt()`. Compute addition, subtraction, multiplication, division, and modulus, and print the results.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 01: Arithmetic Operations
import java.util.Scanner;

public class Q01 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int num1 = sc.nextInt();

        System.out.print("Enter second integer: ");
        int num2 = sc.nextInt();

        int sum = num1 + num2;
        int diff = num1 - num2;
        int product = num1 * num2;

        System.out.println();
        System.out.println("--- Arithmetic Results ---");
        System.out.println("First Number: " + num1);
        System.out.println("Second Number: " + num2);
        System.out.println("Addition: " + sum);
        System.out.println("Subtraction: " + diff);
        System.out.println("Multiplication: " + product);

        if (num2 != 0) {
            int quotient = num1 / num2;
            int remainder = num1 % num2;
            System.out.println("Division: " + quotient);
            System.out.println("Remainder: " + remainder);
        } else {
            System.out.println("Division & Remainder: Cannot divide by zero.");
        }

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter first integer: Enter second integer:
--- Arithmetic Results ---
First Number: 20
Second Number: 6
Addition: 26
Subtraction: 14
Multiplication: 120
Division: 3
Remainder: 2
```

QUESTION 02

Write a Java program to store the radius of a circle in a variable and calculate its area and circumference. Formula:
Area = $\pi \times r \times r$, Circumference = $2 \times \pi \times r$.

LOGIC

Read radius from console using `sc.nextDouble()`. Calculate `area = Math.PI * radius * radius` and `circumference = 2 * Math.PI * radius`.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 02: Area of a Circle
import java.util.Scanner;

public class Q02 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter radius of the circle: ");
        double radius = sc.nextDouble();

        double pi = Math.PI;
        double area = pi * radius * radius;
        double circumference = 2 * pi * radius;

        System.out.println();
        System.out.println("--- Circle Calculations ---");
        System.out.println("Radius: " + radius);
        System.out.println("Area of Circle: " + area);
        System.out.println("Circumference of Circle: " + circumference);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter radius of the circle:
--- Circle Calculations ---
Radius: 7.0
Area of Circle: 153.93804002589985
Circumference of Circle: 43.982297150257104
```

QUESTION 03

Write a Java program to accept principal amount, rate of interest and time using variables and calculate the simple interest and total amount. Formula: Simple Interest = $(P \times R \times T) / 100$, Amount = $P + \text{Simple Interest}$.

LOGIC

Read `principal`, `rate`, and `time` from console using `sc.nextDouble()`. Compute `simpleInterest = (P \* R \* T) / 100` and `totalAmount = principal + simpleInterest`.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 03: Simple Interest
import java.util.Scanner;

public class Q03 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Principal Amount: ");
        double principal = sc.nextDouble();

        System.out.print("Enter Rate of Interest (%): ");
        double rate = sc.nextDouble();

        System.out.print("Enter Time Period (in years): ");
        double time = sc.nextDouble();

        double simpleInterest = (principal * rate * time) / 100;
        double totalAmount = principal + simpleInterest;

        System.out.println();
        System.out.println("--- Simple Interest Calculation ---");
        System.out.println("Principal Amount: Rs. " + principal);
        System.out.println("Rate of Interest: " + rate + "%");
        System.out.println("Time Period: " + time + " years");
        System.out.println("Simple Interest: Rs. " + simpleInterest);
        System.out.println("Total Amount Payable: Rs. " + totalAmount);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter Principal Amount: Enter Rate of Interest (%): Enter Time Period (in years):
--- Simple Interest Calculation ---
Principal Amount: Rs. 10000.0
Rate of Interest: 5.5%
Time Period: 3.0 years
Simple Interest: Rs. 1650.0
Total Amount Payable: Rs. 11650.0
```

QUESTION 04

Write a Java program to convert temperature from Celsius to Fahrenheit. Formula: Fahrenheit = (Celsius × 9 / 5) + 32.

LOGIC

Read Celsius temperature using `sc.nextDouble()`. Convert to Fahrenheit using `(celsius \* 9.0 / 5.0) + 32.0`.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 04: Temperature Conversion
import java.util.Scanner;

public class Q04 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter temperature in Celsius: ");
        double celsius = sc.nextDouble();

        double fahrenheit = (celsius * 9.0 / 5.0) + 32.0;

        System.out.println();
        System.out.println("--- Temperature Conversion ---");
        System.out.println("Celsius: " + celsius + " deg C");
        System.out.println("Fahrenheit: " + fahrenheit + " deg F");

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter temperature in Celsius:
--- Temperature Conversion ---
Celsius: 37.0 deg C
Fahrenheit: 98.6 deg F
```

QUESTION 05

Write a Java program to store marks of three subjects in variables and calculate the total and average marks.

LOGIC

Read marks for three subjects using `sc.nextInt()`. Calculate `total = subject1 + subject2 + subject3` and `average = total / 3.0`.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 05: Calculate Total and Average
import java.util.Scanner;

public class Q05 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter marks for Subject 1: ");
        int subject1 = sc.nextInt();

        System.out.print("Enter marks for Subject 2: ");
        int subject2 = sc.nextInt();

        System.out.print("Enter marks for Subject 3: ");
        int subject3 = sc.nextInt();

        int total = subject1 + subject2 + subject3;
        double average = total / 3.0;

        System.out.println();
        System.out.println("--- Marks Summary ---");
        System.out.println("Subject 1: " + subject1);
        System.out.println("Subject 2: " + subject2);
        System.out.println("Subject 3: " + subject3);
        System.out.println("Total Marks: " + total);
        System.out.println("Average Marks: " + average);

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter marks for Subject 1: Enter marks for Subject 2: Enter marks for Subject 3:
--- Marks Summary ---
Subject 1: 85
Subject 2: 90
Subject 3: 78
Total Marks: 253
Average Marks: 84.33333333333333
```

QUESTION 06

Write a Java program to accept an integer and check whether the number is even or odd using if-else.

LOGIC

Read integer using `sc.nextInt()`. Check `number % 2 == 0`. Print whether EVEN or ODD.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 06: Even or Odd
import java.util.Scanner;

public class Q06 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int number = sc.nextInt();

        if (number % 2 == 0) {
            System.out.println(number + " is an EVEN number.");
        } else {
            System.out.println(number + " is an ODD number.");
        }

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

Enter an integer: 15 is an ODD number.

QUESTION 07

Write a Java program to accept an integer and check whether it is positive, negative, or zero using if and if-else.

LOGIC

Read integer using `sc.nextInt()`. Use `if-else-if` to check if POSITIVE (`> 0`), NEGATIVE (`< 0`), or ZERO (`== 0`).

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 07: Positive, Negative or Zero
import java.util.Scanner;

public class Q07 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int number = sc.nextInt();

        if (number > 0) {
            System.out.println(number + " is POSITIVE.");
        } else if (number < 0) {
            System.out.println(number + " is NEGATIVE.");
        } else {
            System.out.println("The entered number is ZERO.");
        }

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

Enter an integer: -12 is NEGATIVE.

QUESTION 08

Write a Java program to accept two numbers and find the larger number using if-else. If both numbers are equal, display an appropriate message.

LOGIC

Read two integers using `sc.nextInt()`. Compare values and print the larger number or an equality message.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 08: Largest of Two Numbers
import java.util.Scanner;

public class Q08 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int num1 = sc.nextInt();

        System.out.print("Enter second number: ");
        int num2 = sc.nextInt();

        if (num1 > num2) {
            System.out.println(num1 + " is larger than " + num2);
        } else if (num2 > num1) {
            System.out.println(num2 + " is larger than " + num1);
        } else {
            System.out.println("Both numbers are equal.");
        }

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

```
Enter first number: Enter second number: 78 is larger than 45
```

QUESTION 09

Write a Java program to accept a person's age and check whether the person is eligible to vote (Age $\geq$ 18).

LOGIC

Read age using `sc.nextInt()`. Evaluate `age  $\geq$  18` and display voting eligibility status.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 09: Voting Eligibility
import java.util.Scanner;

public class Q09 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter person's age: ");
        int age = sc.nextInt();

        if (age  $\geq$  18) {
            System.out.println("Status: Eligible to vote.");
        } else {
            System.out.println("Status: Not eligible to vote.");
        }

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

Enter person's age: Status: Eligible to vote.

QUESTION 10

Write a Java program to accept marks of a student and check whether the student has passed or failed (Marks >= 40).

LOGIC

Read student marks using `sc.nextInt()`. Evaluate `marks >= 40` and print PASS or FAIL.

CODE (INTERACTIVE SCANNER INPUT)

```
// Assignment 01 - Question 10: Student Result
import java.util.Scanner;

public class Q10 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter student marks: ");
        int marks = sc.nextInt();

        if (marks >= 40) {
            System.out.println("Result: PASS");
        } else {
            System.out.println("Result: FAIL");
        }

        sc.close();
    }
}
```

SAMPLE INTERACTIVE RUN & OUTPUT

Enter student marks: Result: PASS
